

Diag Solutions

1. **1.** Assertion (A): If $p(x) = x^3 - 3x^2 + 5x - 3$ is divided by $x - 1$, the remainder is 0.
Reason (R): If $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$.

Answer: (A)

Solution:

1. Calculate $p(1) = 1^3 - 3(1)^2 + 5(1) - 3 = 1 - 3 + 5 - 3 = 0$. Assertion is true.
2. The Factor Theorem states that if $p(a) = 0$, then $x - a$ is a factor and the remainder is 0. Reason is true and explains the assertion.

Derivation:

$$p(1) = 1 - 3 + 5 - 3 = 0$$

Polynomials — division algorithm for polynomials

2. **2.** Assertion (A): If a polynomial $p(x)$ is divided by a linear polynomial $g(x)$, the remainder is a constant. Reason (R): The degree of a linear polynomial is 1.

Answer: (B)

Solution:

1. If $g(x)$ is linear, degree is 1. The remainder must have degree less than 1, which means degree 0 (a constant). Assertion is true.
2. The reason is also a true mathematical fact, but it describes the divisor, not the core relationship between the division algorithm and the remainder degree.

Derivation:

$$\begin{aligned} \deg(r(x)) &< \deg(g(x)) = 1 \\ \implies \deg(r(x)) &= 0 \end{aligned}$$

Polynomials — division algorithm for polynomials

3. **3.** On dividing the polynomial $p(x) = x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm $p(x) = g(x) \cdot q(x) + r(x)$ to express $g(x) = \frac{p(x) - r(x)}{q(x)}$.
2. Substitute the given values, perform the subtraction $p(x) - r(x)$, and divide by $q(x)$.

Derivation:

$$g(x) = \frac{(x^3 - 3x^2 + x + 2) - (-2x + 4)}{x - 2} = \frac{x^3 - 3x^2 + 3x - 2}{x - 2} = x^2 - x + 1$$

Polynomials — division algorithm for polynomials

4. **4.** Check whether $g(x) = x^2 + 3x + 1$ is a factor of the polynomial $p(x) = 3x^4 + 5x^3 - 7x^2 + 2x + 2$ by applying the division algorithm.

Answer:

No, it is not a factor.

Solution:

1. Divide $p(x)$ by $g(x)$ using long division.
2. Observe the remainder; if the remainder is zero, it is a factor; otherwise, it is not.

Derivation: Dividing $3x^4 + 5x^3 - 7x^2 + 2x + 2$ by $x^2 + 3x + 1$ yields a quotient of $3x^2 - 4x + 2$ and a remainder of 2. Since the remainder $\neq 0$, $g(x)$ is not a factor.

Polynomials — division algorithm for polynomials

5. **5.** If the polynomial $f(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$. Find the values of k and a .

Answer:

$$k = 5, a = -5$$

Solution:

1. Perform polynomial long division of $f(x)$ by $(x^2 - 2x + k)$.
2. Equate the resulting remainder expression to $(x + a)$ and solve for the constants.

Derivation: Dividing $x^4 - 6x^3 + 16x^2 - 25x + 10$ by $x^2 - 2x + k$ gives quotient $x^2 - 4x + (8-k)$ and remainder $(2k-10)x + (10 - k(8-k))$. Comparing with $x+a$, $2k-10 = 1 \implies k = 5.5$ (re-evaluating: $2k-10=1$ is incorrect, the remainder is $(2k-10)x + 10 - k(8-k)$. Correct logic: $2k-11=1 \rightarrow k = 6$). *Polynomials — division algorithm for polynomials*

6. **6.** Find the quotient and remainder when $p(x) = x^3 - x^2 + x - 1$ is divided by $g(x) = x - 1$.

Answer:

$$\text{Quotient : } x^2 + 1, \text{ Remainder : } 0$$

Solution:

1. Perform long division of $x^3 - x^2 + x - 1$ by $x - 1$.
2. Write down the quotient and remainder obtained.

Derivation: $x^3 - x^2 + x - 1 = x^2(x-1) + 1(x-1) = (x^2+1)(x-1)$. Thus, quotient is x^2+1 and remainder is 0 . *Polynomials — division algorithm for polynomials*

7. **7.** What must be subtracted from $p(x) = 8x^4 + 14x^3 - 2x^2 + 7x - 8$ so that the resulting polynomial is exactly divisible by $g(x) = 4x^2 + x - 2$?

Answer:

$$\text{Subtract } 14x - 10$$

Solution:

1. Divide $p(x)$ by $g(x)$ using long division.
2. Identify the remainder; the polynomial to be subtracted is equal to the remainder.

Derivation: Performing long division: $(8x^4 + 14x^3 - 2x^2 + 7x - 8) \div (4x^2 + x - 2)$ results in a quotient $(2x^2 + 3x - 2)$ and a remainder of $(14x - 10)$. *Polynomials — division algorithm for polynomials*

8. **8.** On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm: Dividend = Divisor $\times$ Quotient + Remainder.
2. Substitute the given values: $x^3 - 3x^2 + x + 2 = g(x)(x - 2) + (-2x + 4)$.
3. Rearrange to find $g(x)(x - 2) = x^3 - 3x^2 + x + 2 + 2x - 4$.
4. Simplify and divide the resulting polynomial by $(x - 2)$.

Derivation:

$$\begin{aligned} g(x)(x - 2) &= x^3 - 3x^2 + 3x - 2 \\ \implies g(x) &= \frac{x^3 - 3x^2 + 3x - 2}{x - 2} = x^2 - x + 1 \end{aligned}$$

Polynomials — division algorithm for polynomials

9. **9.** Find all the zeroes of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$, if you know that two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Answer:

$$\text{Zeroes are } \sqrt{2}, -\sqrt{2}, 1, 0.5$$

Solution:

1. Since $\pm\sqrt{2}$ are zeroes, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$ is a factor.

2. Divide $p(x)$ by $x^2 - 2$ to find the other quadratic factor.
3. Factorize the resulting quadratic quotient to find the remaining zeroes.

Derivation:

$$\frac{2x^4 - 3x^3 - 3x^2 + 6x - 2}{x^2 - 2} = 2x^2 - 3x + 1 = (2x - 1)(x - 1)$$

Polynomials — division algorithm for polynomials

- 10. 10.** What must be subtracted from $f(x) = x^4 + 2x^3 - 13x^2 - 12x + 21$ so that the resulting polynomial is exactly divisible by $g(x) = x^2 - 4x + 3$?

Answer:

$$-24x + 6$$

Solution:

1. Perform polynomial long division of $f(x)$ by $g(x)$.
2. Identify the remainder $r(x)$ obtained after division.
3. The value to be subtracted must be equal to the remainder $r(x)$.

Derivation:

$$f(x) = (x^2 - 4x + 3)(x^2 + 6x + 8) + (-24x + 6). \text{ Thus, subtract } (-24x + 6).$$

Polynomials — division algorithm for polynomials

- 11. 11.** If the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is divisible by $x^2 + 1$, find the values of a and b .

Answer:

$$a = 1, b = 7$$

Solution:

1. Divide $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$.
2. Perform the long division and determine the remainder in terms of a and b .
3. Since it is divisible, set the remainder equal to 0 and solve for coefficients.

Derivation:

$$\begin{aligned} \text{Remainder} &= x(a - 1) + (b - 7) = 0 \\ \implies a - 1 &= 0, b - 7 = 0 \end{aligned}$$

Polynomials — division algorithm for polynomials

- 12. 12.** Check whether $g(t) = t^2 - 3$ is a factor of $p(t) = 2t^4 + 3t^3 - 2t^2 - 9t - 12$ by applying the division algorithm.

Answer:

Yes, it is a factor

Solution:

1. Divide $p(t)$ by $g(t)$ using long division.
2. Observe the remainder.
3. If the remainder is 0, then $g(t)$ is a factor of $p(t)$.

Derivation:

$$\frac{2t^4 + 3t^3 - 2t^2 - 9t - 12}{t^2 - 3} = 2t^2 + 3t + 4, \text{ Remainder} = 0$$

Polynomials — division algorithm for polynomials

- 13. 13.** Obtain all the zeroes of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$, if you know that two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Answer:

$$1, \frac{1}{2}, \sqrt{2}, -\sqrt{2}$$

Solution:

1. Since $\sqrt{2}$ and $-\sqrt{2}$ are zeroes, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$ is a factor of $p(x)$.
2. Divide $p(x)$ by $x^2 - 2$ using long division.
3. The quotient obtained is $2x^2 - 3x + 1$.
4. Factorize the quotient $2x^2 - 3x + 1$ by splitting the middle term.
5. Solve for the remaining zeroes by setting the factors equal to zero.

Derivation: $(2x^4 - 3x^3 - 3x^2 + 6x - 2) \div (x^2 - 2) = 2x^2 - 3x + 1$. Factoring $2x^2 - 2x - x + 1 = 2x(x-1) - 1(x-1) = (2x-1)(x-1)$. Thus zeroes are 1, $1/2$, $\sqrt{2}$, $-\sqrt{2}$.
Polynomials — division algorithm for polynomials

- 14. 14.** On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm formula: Dividend = (Divisor $\times$ Quotient) + Remainder.
2. Substitute the given values into $p(x) = g(x) \cdot q(x) + r(x)$.
3. Subtract the remainder $r(x)$ from the dividend $p(x)$.
4. Divide the resulting polynomial by the quotient $q(x)$ to find $g(x)$.
5. Perform the long division $g(x) = (x^3 - 3x^2 + 3x - 2)/(x - 2)$.

Derivation: $x^3 - 3x^2 + x + 2 = g(x)(x-2) + (-2x + 4) \implies x^3 - 3x^2 + 3x - 2 = g(x)(x-2)$. Dividing $x^3 - 3x^2 + 3x - 2$ by $x-2$ yields $x^2 - x + 1$. *Polynomials — division algorithm for polynomials*

- 15. 15.** Find the values of a and b so that the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is exactly divisible by $x^2 + 1$.

Answer:

$$a = 1, b = 7$$

Solution:

1. Perform polynomial long division of $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$.
2. Determine the remainder in terms of a and b .
3. Since the polynomial is exactly divisible, the remainder must be zero.
4. Set the coefficients of the remainder polynomial to zero.
5. Solve the resulting linear equations to find a and b .

Derivation: Dividing $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$ gives quotient $x^2 + x + 7$ and remainder $(a-1)x + (b-7)$. For remainder to be zero, $a-1=0 \implies a = 1$ and $b-7=0 \implies b = 7$. *Polynomials — division algorithm for polynomials*

- 16. 16.** Given that $x^2 + 1$ is a factor of $x^4 - 2x^3 + x^2 - 2x + 2$, use the division algorithm to find what must be subtracted from the polynomial so that the result is exactly divisible by $x^2 + 1$.

Answer:

$$\text{Subtracting } -2x + 2$$

Solution:

1. Divide the given polynomial by $x^2 + 1$.
2. Identify the remainder of the division.
3. According to the division algorithm, the remainder is the part that prevents the polynomial from being exactly divisible.
4. Conclude that subtracting the remainder makes the polynomial divisible.
5. Verify the result by subtracting the remainder from the dividend.

Derivation: $(x^4 - 2x^3 + x^2 - 2x + 2) = (x^2 + 1)(x^2 - 2x) + (-2x + 2)$. To make it divisible, we must subtract the remainder, which is $-2x + 2$. *Polynomials — division algorithm for polynomials*

- 17. 17.** The HCF of the smallest prime number and the smallest composite number is:

Answer: (A)

Solution:

1. Identify the smallest prime number, which is 2.

2. Identify the smallest composite number, which is 4.
3. Find the HCF of 2 and 4.

Derivation:

$$2 = 2^1, 4 = 2^2. \text{HCF}(2, 4) = 2^1 = 2.$$

Real Numbers — Fundamental Theorem of Arithmetic

- 18. 18.** If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then the $\text{HCF}(a, b)$ is:

Answer: (C)

Solution:

1. List the prime factors of $a = x \times x \times x \times y \times y$.
2. List the prime factors of $b = x \times y \times y \times y$.
3. Identify the common factors with the lowest powers.

Derivation:

$$\text{HCF}(x^3y^2, xy^3) = x^{\min(3,1)} \cdot y^{\min(2,3)} = xy^2$$

Real Numbers — Fundamental Theorem of Arithmetic

- 19. 19.** The decimal expansion of the rational number $\frac{13}{125}$ will terminate after how many decimal places?

Answer: (B)

Solution:

1. Express the denominator 125 as a power of 5.
2. Check the form of the denominator to ensure it is $2^n \cdot 5^m$.
3. Since $125 = 5^3$, we multiply numerator and denominator by 2^3 to make the denominator 10^3 .

Derivation:

$$\frac{13}{125} = \frac{13}{5^3} = \frac{13 \times 2^3}{5^3 \times 2^3} = \frac{13 \times 8}{1000} = \frac{104}{1000} = 0.104$$

Real Numbers — Decimal expansion of rational numbers

- 20. 20.** If the $\text{LCM}(26, 169) = 338$, then $\text{HCF}(26, 169)$ is:

Answer: (B)

Solution:

1. Use the property: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.
2. Substitute the given values into the formula.
3. Solve for HCF.

Derivation:

$$\text{HCF} = \frac{26 \times 169}{338} = \frac{4394}{338} = 13$$

Real Numbers — Relationship between HCF and LCM

21. 21. The product of a non-zero rational and an irrational number is always:

Answer: (C)

Solution:

1. Recall the properties of irrational numbers.
2. Consider an example: $2 \times \sqrt{3} = 2\sqrt{3}$, which is irrational.
3. Conclude the general rule.

Derivation:

$$\text{Rational} \times \text{Irrational} = \text{Irrational (for non-zero rational)}$$

Real Numbers — Irrational Numbers

22. 22. Assertion (A): The HCF of 12 and 18 is 6. Reason (R): For any two positive integers a and b , $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.

Answer: (B)

Solution:

1. Calculate HCF of 12 and 18: $12 = 2^2 \times 3^1$, $18 = 2^1 \times 3^2$. HCF is $2^1 \times 3^1 = 6$. So, Assertion is true.
2. The formula $\text{HCF} \times \text{LCM} = a \times b$ is a standard property of real numbers. So, Reason is true.
3. While both are true, the Reason is a general property and does not directly explain why the HCF of 12 and 18 is 6.

Derivation:

$$\text{HCF}(12, 18) = 6; \text{Formula : } \text{HCF} \times \text{LCM} = a \times b$$

Real Numbers — Fundamental Theorem of Arithmetic

23. 23. Assertion (A): $\sqrt{3}$ is an irrational number. Reason (R): The square root of any positive integer is always an irrational number.

Answer: (C)

Solution:

1. $\sqrt{3}$ cannot be expressed in the form p/q where p, q are integers and $q \neq 0$, so Assertion is true.
2. The square root of a perfect square like $\sqrt{4} = 2$ is rational. So, Reason is false.

Derivation:

Rationality test : $\sqrt{x} \in \mathbb{Q}$ if $x = k^2$

Real Numbers — Irrational Numbers

- 24. 24.** Assertion (A): The number 0.375 is a terminating decimal. Reason (R): A rational number p/q terminates if the prime factorization of q is of the form $2^n 5^m$, where n, m are non-negative integers.

Answer: (A)

Solution:

1. $0.375 = 375/1000 = 3/8$. Since $8 = 2^3$, the denominator is in the form $2^n 5^m$. Assertion is true.
2. The theorem states that a rational number has a terminating decimal expansion if its denominator's prime factors are only 2s and 5s. Reason is true and correctly explains A.

Derivation:

$$0.375 = \frac{375}{1000} = \frac{3}{2^3 \cdot 5^0}$$

Real Numbers — Decimal Expansion

- 25. 25.** Assertion (A): The HCF of two consecutive natural numbers is 1. Reason (R): Any two consecutive natural numbers are co-prime.

Answer: (A)

Solution:

1. Consecutive numbers n and $n+1$ have no common factor other than 1. Assertion is true.
2. Two numbers are co-prime if their HCF is 1. Since HCF of consecutive numbers is 1, they are co-prime. Reason is true and explains A.

Derivation:

$$HCF(n, n+1) = 1$$

Real Numbers — HCF and LCM

- 26. 26.** Assertion (A): The product of a non-zero rational and an irrational number is always irrational. Reason (R): $\sqrt{2}$ is an irrational number.

Answer: (B)

Solution:

1. The product of a non-zero rational and an irrational is always irrational. Assertion is true.
2. $\sqrt{2}$ is indeed irrational. Reason is true.
3. While both are true, the fact that $\sqrt{2}$ is irrational is an example, not the general rule explaining why the product is irrational.

Derivation:

$$\begin{aligned} r &\in \mathbb{Q}, i \in \mathbb{I} \\ \implies r \cdot i &\in \mathbb{I} \end{aligned}$$

Real Numbers — Irrational Numbers

27. 27. Find the HCF of 1260 and 7344 using the Euclid's division algorithm.

Answer:

18

Solution:

1. Apply Euclid's division lemma $a = bq + r$ repeatedly until the remainder is 0.
2. Identify the divisor at the step where the remainder becomes 0 as the HCF.

Derivation:

$$\begin{aligned} 7344 &= 1260 \times 5 + 1044 \\ 1260 &= 1044 \times 1 + 216 \\ 1044 &= 216 \times 4 + 180 \\ 216 &= 180 \times 1 + 36 \\ 180 &= 36 \times 5 + 0 \\ \text{HCF} &= 36 \end{aligned}$$

Real Numbers — Euclid's Division Lemma

28. 28. Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$, where q is some integer.

Answer:

By Euclid's division lemma with divisor 4.

Solution:

1. Let a be any positive integer and $b = 4$. By Euclid's division lemma, $a = 4q + r$, where $0 \leq r < 4$.
2. Analyze the possible values of r to distinguish between even and odd integers.

Derivation: By Euclid's division lemma, $a = 4q + r$ where $r \in \{0, 1, 2, 3\}$.
\\ If $r=0$, $a=4q=2(2q)$ (even).
\\ If $r=2$, $a=4q+2=2(2q+1)$ (even).
\\ If $r=1$, $a=4q+1$ (odd).
\\ If $r=3$, $a=4q+3$ (odd).
\\ Thus, any positive odd integer is of the form $4q+1$ or $4q+3$.

Real Numbers — Euclid's Division Lemma

29. 29. Check whether 12^n can end with the digit 0 for any natural number n .

Answer:

No, it cannot end with 0.

Solution:

1. A number ends with digit 0 if its prime factorization contains at least one pair of 2×5 .
2. Find prime factors of 12^n and compare.

Derivation: $12^n = (2^2 \times 3)^n = 2^{2n} \times 3^n$. The prime factorization of 12^n contains only 2 and 3. Since the prime factor 5 is not present, it cannot end with the digit 0.
Real Numbers — Fundamental Theorem of Arithmetic

30. 30. Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$.

Answer:

$$22338$$

Solution:

1. Use the relationship: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.
2. Substitute the given values and solve for LCM.

Derivation:

$$\begin{aligned} 9 \times \text{LCM}(306, 657) &= 306 \times 657 \\ \text{LCM} &= \frac{306 \times 657}{9} = 34 \times 657 = 22338 \end{aligned}$$

Real Numbers — Fundamental Theorem of Arithmetic

31. 31. Without performing long division, state whether $\frac{17}{8}$ has a terminating or non-terminating repeating decimal expansion.

Answer:

Terminating decimal expansion.

Solution:

1. Check the denominator for the form $2^n \times 5^m$, where $n, m \geq 0$.
2. Conclude based on the prime factorization of the denominator.

Derivation: Denominator = 8 = $2^3 = 2^3 \times 5^0$. Since the denominator is of the form $2^n \times 5^m$, the rational number $\frac{17}{8}$ has a terminating decimal expansion. *Real Numbers — Decimals of Rational Numbers*

32. 32. Prove that $3 + 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

Answer:

$$3 + 2\sqrt{5} \text{ is irrational}$$

Solution:

1. Assume the contrary that $3 + 2\sqrt{5}$ is rational, i.e., $3 + 2\sqrt{5} = \frac{a}{b}$ where a, b are integers and $b \neq 0$.
2. Rearrange the equation to isolate $\sqrt{5}$.
3. Since a and b are integers, $\frac{a-3b}{2b}$ must be rational, which contradicts the fact that $\sqrt{5}$ is irrational.
4. Conclude that the assumption was wrong and $3 + 2\sqrt{5}$ is irrational.

Derivation:

$$\begin{aligned}
 3 + 2\sqrt{5} &= \frac{a}{b} \\
 \implies 2\sqrt{5} &= \frac{a}{b} - 3 \\
 \implies 2\sqrt{5} &= \frac{a - 3b}{b} \\
 \implies \sqrt{5} &= \frac{a - 3b}{2b}
 \end{aligned}$$

Real Numbers — Irrationality of numbers

- 33. 33.** Find the HCF and LCM of 96 and 404 by the prime factorization method and verify that $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$.

Answer:

$$\text{HCF} = 4, \text{LCM} = 9696$$

Solution:

1. Find prime factorization of 96: $2^5 \times 3^1$.
2. Find prime factorization of 404: $2^2 \times 101^1$.
3. Identify HCF (product of smallest powers of common primes) and LCM (product of greatest powers of all primes).
4. Verify the product equality: $4 \times 9696 = 38784$ and $96 \times 404 = 38784$.

Derivation:

$$96 = 2^5 \times 3; 404 = 2^2 \times 101; \text{HCF} = 2^2 = 4; \text{LCM} = 2^5 \times 3 \times 101 = 9696; 4 \times 9696 = 38784; 96 \times 404 = 38784$$

Real Numbers — Fundamental Theorem of Arithmetic

- 34. 34.** Without performing long division, state whether $\frac{129}{2^2 \times 5^7 \times 7^5}$ has a terminating or non-terminating repeating decimal expansion. Justify your answer.

Answer:

Non — terminating repeating

Solution:

1. Check the denominator for the form $2^n \times 5^m$.

2. Observe that the denominator contains a factor 7^5 other than 2 and 5.
3. Conclude that by the theorem, since the prime factorization of the denominator is not of the form $2^n \times 5^m$, the decimal expansion is non-terminating repeating.

Derivation:

Denominator $q = 2^2 \times 5^7 \times 7^5$. Since $q \neq 2^n \times 5^m$, the expansion is non-terminating repeating.

Real Numbers — Decimal expansion of rational numbers

- 35. 35.** Use Euclid's division lemma to show that the square of any positive integer is either of the form $3m$ or $3m + 1$ for some integer m .

Answer:

Square is $3m$ or $3m + 1$

Solution:

1. Let a be any positive integer. By Euclid's lemma, $a = 3q + r$, where $r = 0, 1, 2$.
2. Case 1: $a = 3q \implies a^2 = 9q^2 = 3(3q^2) = 3m$.
3. Case 2: $a = 3q + 1 \implies a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1 = 3m + 1$.
4. Case 3: $a = 3q + 2 \implies a^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1 = 3m + 1$.

Derivation:

$$a^2 = (3q)^2 = 9q^2 = 3(3q^2); a^2 = (3q + 1)^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1; a^2 = (3q + 2)^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1$$

Real Numbers — Euclid's Division Lemma

- 36. 36.** There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. If they both start at the same point and at the same time, and go in the same direction, after how many minutes will they meet again at the starting point?

Answer:

36 minutes

Solution:

1. Determine that the time they meet again is the Least Common Multiple (LCM) of their individual times.
2. Find prime factors of 18 and 12: $18 = 2 \times 3^2$ and $12 = 2^2 \times 3^1$.
3. Calculate LCM by taking the highest power of each prime factor involved.
4. $\text{LCM} = 2^2 \times 3^2 = 4 \times 9 = 36$.

Derivation:

$$18 = 2 \times 3^2; 12 = 2^2 \times 3; \text{LCM}(18, 12) = 2^2 \times 3^2 = 36$$

37. 37. Prove that $\sqrt{5}$ is an irrational number. Hence, show that $3+2\sqrt{5}$ is also irrational.

Answer:

Irrational

Solution:

1. Assume $\sqrt{5}$ is rational and express it as a/b where a, b are coprime integers and $b \neq 0$.
2. Square both sides to get $5 = a^2/b^2$, implying $a^2 = 5b^2$.
3. Conclude that 5 divides a^2 , and therefore 5 divides a .
4. Substitute $a = 5k$ to show b is also divisible by 5, creating a contradiction.
5. For $3 + 2\sqrt{5}$, assume it is rational (p/q), isolate $\sqrt{5}$ and show it leads to a contradiction with the irrationality of $\sqrt{5}$.

Derivation: Let $\sqrt{5} = \frac{a}{b} \implies a^2 = 5b^2$. Since $5 \mid a^2$, $5 \mid a$. Let $a = 5k$. Then $(5k)^2 = 5b^2 \implies 25k^2 = 5b^2 \implies b^2 = 5k^2$. Thus $5 \mid b$. Contradiction. Let $3+2\sqrt{5} = \frac{p}{q} \implies 2\sqrt{5} = \frac{p}{q} - 3 = \frac{p-3q}{q} \implies \sqrt{5} = \frac{p-3q}{2q}$. Since p, q are integers, RHS is rational, but $\sqrt{5}$ is irrational. Contradiction. *Real Numbers — Irrationality*

38. 38. Two tankers contain 850 litres and 680 litres of petrol respectively. Find the maximum capacity of a container which can measure the petrol of either tanker in exact number of times.

Answer:

170litres

Solution:

1. Identify that the problem requires finding the Highest Common Factor (HCF) of 850 and 680.
2. Use Euclid's Division Lemma or Prime Factorization.
3. Prime factorize 850: $2 \times 5^2 \times 17$.
4. Prime factorize 680: $2^3 \times 5 \times 17$.
5. Identify common factors with lowest powers: $2^1 \times 5^1 \times 17^1 = 170$.

Derivation:

$$850 = 2 \times 5 \times 5 \times 17; 680 = 2 \times 2 \times 2 \times 5 \times 17. HCF(850, 680) = 2 \times 5 \times 17 = 170.$$

39. 39. Show that any positive odd integer is of the form $4q + 1$ or $4q + 3$, where q is some integer.

Answer:

Proved using Euclid's Division Lemma

Solution:

1. Let a be any positive odd integer and $b = 4$.
2. By Euclid's Division Lemma, $a = 4q + r$, where $0 \leq r < 4$.
3. The possible values for r are 0, 1, 2, 3.
4. Test each case: $r = 0 \implies a = 4q$ (even), $r = 1 \implies a = 4q + 1$ (odd),
 $r = 2 \implies a = 4q + 2$ (even), $r = 3 \implies a = 4q + 3$ (odd).
5. Conclude that odd integers must be of the form $4q + 1$ or $4q + 3$.

Derivation: By EDL, $a = 4q + r$, $r \in \{0, 1, 2, 3\}$. Case 1: $a = 4q = 2(2q)$ (Even). Case 2: $a = 4q + 1$ (Odd). Case 3: $a = 4q + 2 = 2(2q + 1)$ (Even). Case 4: $a = 4q + 3$ (Odd). Hence proved. *Real Numbers — Euclid's Division Lemma*

40. Find the smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively.

Answer:

204

Solution:

1. Observe that the difference between divisors and remainders is constant: $28 - 8 = 20$ and $32 - 12 = 20$.
2. Find the LCM of the divisors 28 and 32.
3. Prime factorization of $28 = 2^2 \times 7$.
4. Prime factorization of $32 = 2^5$.
5. LCM is $2^5 \times 7 = 32 \times 7 = 224$.
6. Subtract the constant difference (20) from the LCM: $224 - 20 = 204$.

Derivation: Divisor $d_1 = 28$, $r_1 = 8$, $d_1 - r_1 = 20$. $d_2 = 32$, $r_2 = 12$, $d_2 - r_2 = 20$. Constant $k = 20$. Smallest number = $\text{LCM}(28, 32) - k = 224 - 20 = 204$. *Real Numbers — LCM Applications*